

SANJIV IYER

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EDUCATION

The University of Texas at Austin — Austin, TX

May 2027

B.S. in Mechanical Engineering; Minor: Semiconductor Manufacturing; Minor: Entrepreneurship

University Honors: Fall 2023, Fall 2025 | GPA: 3.62/4.0

Relevant Coursework: IC Nanomanufacturing Techniques; High Throughput Nanopattern; Chem Engr Nanofab; Mechatronics; Dynamic Systems and Controls; Elements of Computers and Programming; Machine Elements; Heat Transfer; Fluid Mechanics

PROFESSIONAL EXPERIENCE

Samsung Austin Semiconductor — Metals MOL Equipment Engineering Intern; Austin/Taylor, TX

May 2026 — August 2026

- Implemented KLA vibration-sensing wafer across Metals PVD/CVD/ALD mainframes to detect robot hardware degradation before wafer defects occurred; established Statistical Process Control (SPC) limits on vibration data to proactively flag tool owners of robot faults
- Developed and deployed a Failure Detection Control (FDC) app enabling equipment engineers to identify top interlocking parameters in a single click, cutting trend-analysis time 3X versus the legacy system; surfaced 10+ trends that prevented over \$1M in interlock stop loss/wafer scrap
- Designed a preclean chamber cover to resolve particle contamination issues on Metals PVD chambers during preventative maintenance, preventing an estimated \$250K in wafer scrap loss; produced technical drawings and manufactured 4 covers in PETG and Acetal
- Implemented an IoT DAQ system to integrate secondary sensors (thermocouples, limit switches, vibration sensors) into the fab network using Power over Ethernet (PoE); established a platform for future sensor hookups and continuous tool health monitoring where none previously existed

Marlinspike — Venture Capital Analyst Intern; Washington, D.C.

January 2026 — May 2026

- Produced 25+ investment screening memos on early-stage aerospace, defense, and energy/semiconductor startups; evaluated product differentiation, technical feasibility, and market opportunity through due diligence
- Built financial models incorporating market capture, dilution, and cash flow projections to support investment committee recommendations
- Assessed technical feasibility and go-to-market momentum; flagged key risks in bi-weekly presentations that led to investment decisions

Fisher Energy — Mechanical Engineering Intern; Houston, TX

May 2025 — August 2025

- Owned design, modeling, and manufacturing of 7 unique RNG pressure vessels in high-pressure/temperature environments
- Led redesign process of davit arms, increasing profit margin over 60% through material selection and manufacturing technique
- Created and revised over 40 3D and P&ID drawings with GD&T and weld symbols for clear manufacturing; developed an Excel-based coalescing filter sizing datasheet that reduced client turnaround times by 75%

Crean — Engineering Consulting Intern; Austin, TX

April 2024 — August 2024

- Owned design and drafting of 8 subassemblies in an automated multi-axis antenna positioning system used for RF wave testing; converted pulley mechanisms to lead screws to improve positioning precision for accurate signal measurement
- Created SOLIDWORKS parts and assemblies (SS, Al, PETG, Delrin) with full part drawings and exploded assembly views
- Supported vendor selection, BOM creation, and quoting; applied GD&T principles to resolve shaft misalignment and clearance interference issues

ACTIVITIES

Texas Rocket Engineering Lab (TREL) — Airframe Manufacturing Engineer; Austin, TX

September 2023 — August 2024

- Designed and fabricated carbon fiber rocket skirts and structural integration with COPV tank; included the design and manufacturing of a stainless-steel custom alignment fixture that standardized layup process and halved production time
- Identified bolt failure during tensile testing of skirt joints; proposed, modeled, and prototyped a new flange-style interface for skirt connection
- Instrumented/built LOX tank with strain gauges and supported bonded joint wet layups, COPV blanket pressure testing, and integration

ASME — Member; Austin, TX

August 2023 — Present

- Collaborating in an engineering professional organization that enables skill and professional development through technical talks and workshops
- Networking monthly with larger corporations in fields such as energy and aerospace; gaining exposure to technical trends and career paths

FTC Team Phantom #12857 — Team Captain; Magnolia, TX

Summer 2018 — March 2022

- Led an elite First Tech Challenge team to the World Championships twice, practiced leadership and compassion throughout collaboration
- Recognized and awarded by Congressman Kevin Brady for spreading STEM throughout the community; helped start/mentor 2 junior teams

LEADERSHIP AND COMMUNITY INVOLVEMENT

Texas Wranglers — Citywide Chairman and Member; Austin, TX

August 2023 — Present

- Raised \$25,000 toward the organization's \$475,000 total raised over 27 years for Easterseals Greenleaf Texas; directly impacted over 15 veterans
- Led citywide service efforts and represented the organization in leadership meetings and university-wide service councils

National Charity Roundtable — President and Community Volunteer; The Woodlands, TX

Summer 2017 — May 2023

- Volunteered 75+ hours annually to serve the greater Woodlands community; worked with 40+ philanthropies
- Delivered 100+ meals to senior citizens annually and built lasting connections with local businesses such as Meals on Wheels

ADDITIONAL INFORMATION

Computer Skills: SOLIDWORKS, Python, Inventor, Fusion 360, AutoCAD, MATLAB, Java, Excel | **Languages:** Fluent in English and Tamil

Interests: Dad Rock, Formula 1, Guitar, Space, Tennis, Football, and Soccer | **Work Eligibility:** Eligible to work in the U.S. with no restrictions